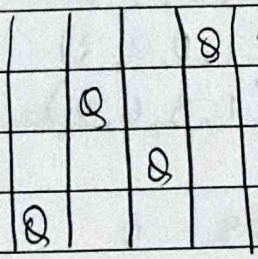
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=> Hill Climbing Algorithm

Start
For each iteration :
generate all neighbouring states
Set best_state = current_state
For each neighbour :
If value (neighbour) > value (best_state)
best_state = neighbour
If best_state = current_state
break
current_state = best_state
Print current_state.



$(x_0, x_1, x_2, x_3) = (3, 1, 2, 0)$
cost = 2

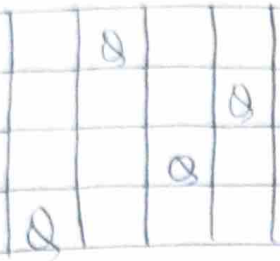
Calculate all neighbouring states.

Swapped Queens	Neighbor state	cost
x_0, x_1	(1, 3, 2, 0)	1
x_0, x_2	(2, 1, 3, 0)	1
x_0, x_3	(0, 1, 2, 3)	6
x_1, x_2	(3, 2, 1, 0)	6
x_1, x_3	(3, 0, 2, 1)	1
x_2, x_3	(3, 1, 0, 2)	1

Minimum cost = 1
(x_0, x_1)

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new current state



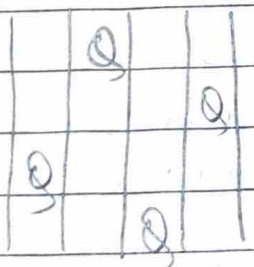
x_1 & x_2 attack each other

current cost = 1

Swapped Queens	Neighbor state	cost
x_0, x_1	(3, 1, 2, 0)	2
x_0, x_2	(2, 3, 1, 0)	2
x_0, x_3	(0, 3, 2, 1)	4
x_1, x_2	(1, 2, 3, 0)	4
x_1, x_3	(1, 0, 2, 3)	2
x_2, x_3	(1, 3, 0, 2)	0

Minimum cost is 0

New current state: (1, 3, 0, 2)



Goal state

$(x_0, x_1, x_2, x_3) = (1, 3, 0, 2)$

Final cost = 0

Iteration	Current state	cost	Selected Swap	Next state
1	(3, 1, 2, 0)	2	x_0, x_1	(1, 3, 2, 0)
2	(1, 3, 2, 0)	1	x_2, x_3	(1, 3, 0, 2)
3	(1, 3, 0, 2)	0		Goal reached